

"Given v and p in $\mathbb{R}^2$ or $\mathbb{R}^3$, the effect of adding p to v is to move v in a direction parallel ~~through~~ to the line through p and O ."

$\therefore$ The given statement is True $\square$

e) The solution set of $Ax=b$ is obtained by translating the solution set of $Ax=0$.

Solution: The given statement is True. This is mentioned in the paragraph just below Theorem 6 on Page 47.

25) Prove the second part of Theorem 6: Let w be any solution of $Ax=b$, and define $v_h = w - p$. Show that v_h is a solution of $Ax=0$. This shows that every solution of $Ax=b$ has the form $w = p + v_h$, with p a particular solution of $Ax=b$ and v_h a solution of $Ax=0$.

Solution: $\because w$ is a solution of $Ax=b$, $Aw=b$. p is a particular solution of $Ax=b$. $\therefore Ap=b$.

$$\begin{aligned} \text{Given, } v_h = w - p. \therefore Av_h &= A(w - p) = A(w + (-1) \cdot p) \quad (\text{By defn.}) \\ &= Aw + A((-1) \cdot p) \quad (\text{Theorem 5(a)}) \\ &= Aw + (-1) \cdot (Ap) \quad (\text{Theorem 5(b)}) \\ &= b + (-1) \cdot b \quad (\text{Given}) = b + (-b) \quad (\text{By Defn.}) \\ &= 0 \quad (\text{Algebraic Property (iv)}) \end{aligned}$$

By Defn, v_h is a solution of $Ax=0$ $\square$

26) Suppose $Ax=b$ has a solution. Explain why the solution is unique precisely when $Ax=0$ has only the trivial solution.

Solution: By Theorem 6, Every solution of $Ax=b$ has the form $w = p + v_h$ with p a particular solution of $Ax=b$ and v_h a solution of $Ax=0$.
Given, $v_h=0$ is the only solution of $Ax=0$, i.e. it has only the trivial solution. $\therefore w = p + v_h = p + 0 = p$ (By Algebraic property (iii))
 $\therefore Ax=b$ has ^{only} the unique solution $w=p$ $\square$